

Yaru Niu

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Education

Carnegie Mellon University

Ph.D. in Mechanical Engineering, Safe AI Lab

Advisor: Prof. Ding Zhao

Pittsburgh, PA

Aug. 2022 – Present

Georgia Institute of Technology

M.S. in Electrical and Computer Engineering, CORE Robotics Lab

Advisor: Prof. Matthew Gombolay

Atlanta, GA

Aug. 2019 – Apr. 2022

South China University of Technology (SCUT)

B.Eng. in Intelligence Science and Technology

Guangzhou, China

Sep. 2015 – Jun. 2019

Publications

(* indicates equal contributions, [Google Scholar](https://scholar.google.com/))

1. **Yaru Niu**, Zhenlong Fang, Binghong Chen, Shuai Zhou, Revanth Senthilkumaran, Hao Zhang, Bingqing Chen, Chen Qiu, H. Eric Tseng, Jonathan Francis, Ding Zhao. Learning Versatile Humanoid Manipulation with Touch Dreaming. *International Conference on Intelligent Robots and Systems (IROS)*, 2026.
2. Hao Zhang, **Yaru Niu**, Yikai Wang, Ding Zhao, H. Eric Tseng. HALyPO: Heterogeneous-Agent Lyapunov Policy Optimization for Human-Robot Collaboration. *International Conference on Machine Learning (ICML)*, 2026 (**Oral**).
3. Dongsu Lee, Daehee Lee, **Yaru Niu**, Honguk Woo, Amy Zhang, Ding Zhao. Team Coordination by Interactive World Latent. *ARLET Workshop at Neural Information Processing Systems (NeurIPS)*, 2025 (**Oral**).
4. **Yaru Niu***, Yunzhe Zhang*, Mingyang Yu, Changyi Lin, Chenhao Li, Yikai Wang, Yuxiang Yang, Wenhao Yu, Tingnan Zhang, Zhenzhen Li, Jonathan Francis, Bingqing Chen, Jie Tan, Ding Zhao. Human2LocoMan: Learning Versatile Quadrupedal Manipulation with Human Pre-Training. *Robotics: Science and Systems (RSS)*, 2025.
5. Yuyou Zhang, Yihang Yao, Shiqi Liu, **Yaru Niu**, Changyi Lin, Yuxiang Yang, Wenhao Yu, Tingnan Zhang, Jie Tan, Ding Zhao. QuietPaw: Learning Quadrupedal Locomotion with Versatile Noise Preference Alignment. *International Conference on Intelligent Robots and Systems (IROS)*, 2025.
6. Yuming Feng*, Chuye Hong*, **Yaru Niu***, Shiqi Liu, Yuxiang Yang, Wenhao Yu, Tingnan Zhang, Jie Tan, Ding Zhao. Learning Multi-Agent Loco-Manipulation for Long-Horizon Quadrupedal Pushing. *International Conference on Robotics and Automation (ICRA)*, 2025.
7. Changyi Lin, Xingyu Liu, Yuxiang Yang, **Yaru Niu**, Wenhao Yu, Tingnan Zhang, Jie Tan, Byron Boots, Ding Zhao. LocoMan: Advancing Versatile Quadrupedal Dexterity with Lightweight Loco-Manipulators. *International Conference on Intelligent Robots and Systems (IROS)*, 2024.
8. Yuyou Zhang, **Yaru Niu**, Xingyu Liu, Ding Zhao. COMPOSER: Scalable and Robust Modular Policies for Snake Robots. *International Conference on Robotics and Automation (ICRA)*, 2024.
9. Haohong Lin, Wenhao Ding, Zuxin Liu, **Yaru Niu**, Jiacheng Zhu, Yuming Niu, Ding Zhao. Safety-aware Causal Representation for Trustworthy Reinforcement Learning in Autonomous Driving. *Robotics and Automation Letters (RA-L)*, 2024.
10. **Yaru Niu**, Shiyu Jin*, Zeqing Zhang*, Jiacheng Zhu, Ding Zhao, Liangjun Zhang. GOATS: Goal Sampling Adaptation for Scooping with Curriculum Reinforcement Learning. *International Conference on Intelligent Robots and Systems (IROS)*, 2023.
11. Mengdi Xu, Peide Huang, **Yaru Niu**, Visak Kumar, Jieli Qiu, Chao Fang, Kuan-Hui Lee, Xuewei Qi, Henry Lam, Bo Li, Ding Zhao. Group Distributionally Robust Reinforcement Learning with Hierarchical Latent Variables. *International Conference on Artificial Intelligence and Statistics (AISTATS)*, 2023.
12. Lingfeng Sun*, Chen Tang*, **Yaru Niu**, Enna Sachdeva, Chiho Choi, Teruhisa Misu, Masayoshi Tomizuka, Wei Zhan. Domain Knowledge Driven Pseudo Labels for Interpretable Goal-conditioned Interactive Trajectory Prediction. *International Conference on Intelligent Robots and Systems (IROS)*, 2022.
13. **Yaru Niu**. Adaptable and Scalable Multi-Agent Graph-Attention Communication. *Master's Thesis, Georgia Institute of Technology*, 2022.

14. Rohan Paleja*, **Yaru Niu***, Andrew Silva, Chace Ritchie, Sugju Choi, Matthew Gombolay. Learning Interpretable, High-Performing Policies for Autonomous Driving. *Robotics: Science and Systems (RSS)*, 2022.
15. **Yaru Niu***, Rohan Paleja*, Matthew Gombolay. Multi-Agent Graph-Attention Communication and Teaming. *International Conference on Autonomous Agents and Multiagent Systems (AAMAS)*, 2021 (**Oral**).
16. **Yaru Niu***, Rohan Paleja*, Matthew Gombolay. MAGIC: Multi-Agent Graph-Attention Communication. *Mair2 Workshop at International Conference on Computer Vision (ICCV)*, 2021 (**Best Paper Award**).
17. Zhijun Zhang*, **Yaru Niu***, Ziyi Yan, Shuyang Lin. Real-time Whole-body Imitation by Humanoid Robots and Task-oriented Teleoperation Using an Analytical Mapping Method and Quantitative Evaluation. *Applied Sciences*, 2018.

Preprints

(* indicates equal contributions)

1. **Yaru Niu***, Binghong Chen*, Changwei Yao*, Shangtao Li*, Zhenlong Fang, Revanth Krishna Senthilkumaran, Shuai Zhou, Yuemin Mao, Kalie Ching, Austin Chao, Yudong Luo, Peiqi Yu, Heng Zhang, Hao Zhang, Bingqing Chen, Chen Qiu, Eric H. Tseng, Changliu Liu, Jonathan Francis, Ding Zhao. EgoX: Egocentric Cross-Embodiment Manipulation with Embodiment Dreaming. *In submission*, 2026.
2. Mohamed Elmoghany, Kehan Li, Ruoqu Chen, Tarun Chiruvolu, Shutong Zhang, Ryan Wang, Stephen Tian, **Yaru Niu**, Michelle Guo, Ding Zhao, Weiyu Liu, Mengdi Xu, Li Fei-Fei, Jiajun Wu. ToolDex: An Ego-Exo Human Demonstration Dataset for Creative Physical Tool Use. *In submission*, 2026.
3. Hao Zhang*, Ruize Geng, Yisen Li, **Yaru Niu**, Yikai Wang, Raihan Haque, Khalil Zibss, Guanyang Luo, Hui-ping Wang, Hongtei (Eric) Tseng, Ding Zhao. Anticipatory Robot Goalkeeping via Monotone Optimal Stopping. *In submission*, 2026.
4. Hongyi Chen, Tony Dong, Tiancheng Wu, Liquan Wang, Yash Jangir, **Yaru Niu**, Yufei Ye, Homanga Bharadhwaj, Zackory Erickson, Jeffrey Ichnowski. Dexterous Manipulation Policies from RGB Human Videos via 3D Hand-Object Trajectory Reconstruction. *In submission*, 2026.
5. Dongsu Lee*, **Yaru Niu***, Yongkang Liu, Ding Zhao. Vector-quantized Graph-attention Transformer for Multi-agent Communication. *In submission*, 2025.
6. Rohan Paleja*, Letian Chen*, **Yaru Niu***, Andrew Silva, Zhaoxin Li, Songan Zhang, Chace Ritchie, Sugju Choi, Kimberlee Chestnut Chang, Hongtei Eric Tseng, Yan Wang, Subramanya Nagesh Rao, Matthew Gombolay. Learning Interpretable, High-Performing Policies for Continuous Control. Submitted to *Journal of Machine Learning Research (JMLR)*, 2023.
7. Mengdi Xu*, Peide Huang*, Wenhao Yu, Shiqi Liu, Xilun Zhang, **Yaru Niu**, Tingnan Zhang, Fei Xia, Jie Tan, Ding Zhao. Creative Robot Tool Use with Large Language Models. *arXiv Preprint*, 2023.
8. Lingdong Kong, Shaoyuan Xie, Hanjiang Hu, **Yaru Niu**, et al. The RoboDrive Challenge: Drive Anytime Anywhere in Any Condition. *arXiv Preprint*, 2024.
9. Lingdong Kong, **Yaru Niu**, Shaoyuan Xie, Hanjiang Hu, et al. The RoboDepth Challenge: Methods and Advancements Towards Robust Depth Estimation. *arXiv Preprint*, 2023.
10. Zhijun Zhang (PI), Lingdong Kong, **Yaru Niu**, Ziyang Liang. Modification of Gesture-Determined-Dynamic Function with Consideration of Margins for Motion Planning of Humanoid Robots. *arXiv Preprint*, 2020.

Patents

1. Zhijun Zhang, **Yaru Niu**. A Mapping Method of Human Postures Applied to Motion Imitation by Humanoid Robots (Translated from Chinese). *Published Authorization Number: CN107953331B*.
2. Zhijun Zhang, **Yaru Niu**. A Similarity Evaluation Method of Imitation by Humanoid Robots (Translated from Chinese). *Published Authorization Number: CN107818318B*.
3. Zhijun Zhang, **Yaru Niu**, Hao Wang. A Mapping Method of Human Body's Rotation and Displacement Applied to Humanoid Robots (Translated from Chinese). *Published Authorization Number: CN108858188B*.
4. Zhijun Zhang, **Yaru Niu**, Hao Wang. An Evaluation Metric of Humanoid Robot and Human Posture Similarity (Translated from Chinese). *Published Application Number: CN109064486A*.

Experience

NVIDIA Research <i>Research Intern, Learning and Perception Research Group</i>	May 2026 – Present <i>Mentors: Ye Yuan, Umar Iqbal</i>
Bosch Center for Artificial Intelligence <i>Machine Learning Research Intern, Robot Learning Lab</i>	May 2025 – Apr. 2026 <i>Mentors: Bingqing Chen, Jonathan Francis</i>
Baidu Research <i>Research Intern, Robotics and Autonomous Driving Lab</i>	Jan. 2022 – Jul. 2022 <i>Mentor: Liangjun Zhang</i>
University of California, Berkeley <i>Research Intern, Mechanical Systems Control Lab</i>	Jul. 2021 – Feb. 2022 <i>Advisor: Masayoshi Tomizuka</i>
South China University of Technology <i>Undergraduate Researcher, Bionic Intelligent Robot Lab</i>	Aug. 2016 – Jun. 2019 <i>Advisor: Zhijun Zhang</i>
University of California, Irvine <i>Undergraduate Researcher, Advanced Integrated Cyber-Physical Systems Lab</i>	Summer 2018 <i>Advisor: Al Faruque</i>

Honors & Awards

Georgia Tech ECE Fellowship	Feb. 2022
Best Paper Award, ICCV 2021 Mair2 Workshop	Oct. 2021
AAMAS 2021 Scholarship	Feb. 2021
National Motivational Scholarship (twice) , awarded by Ministry of Edu. of China (rank 2/51)	Nov. 2016, 2018
The Jetta Scholarship , awarded by Jetta Company Limited (rank 3/51)	Dec. 2017
SCUT Scholarship , awarded by SCUT (rank 3/51)	Nov. 2017
2nd Prize in China Undergrad. Math. Contest in Modeling (CUMCM) (top 5% in SCUT)	Oct. 2017
1st Prize of Guangdong Province in CUMCM (top 5% in SCUT)	Oct. 2017

Teaching Experience

24-784 Trustworthy Artificial Intelligence <i>Graduate Teaching Assistant, Carnegie Mellon University</i>	Jan. 2024 – May 2024
CS 4731/7632 Game Artificial Intelligence <i>Graduate Teaching Assistant, Georgia Institute of Technology</i>	Jun. 2020 – Dec. 2020
CS 4641 Machine Learning <i>Graduate Teaching Assistant, Georgia Institute of Technology</i>	Jan. 2021 – May 2021

Mentorship

Revanth Senthilkumaran (MS at CMU)	Aug. 2025 – Present
Binghong Chen (MS at CMU)	Aug. 2025 – Present
Shuai Zhou (Visiting Researcher at CMU → MS at CMU)	Aug. 2025 – Present
Zhenlong Fang (Visiting Researcher at CMU → PhD at Berkeley)	Aug. 2025 – Present
Ryan Wang (Undergrad. at CMU)	Aug. 2025 – Present
Chenhao Li (MS at CMU)	Jan. 2025 – Jun. 2025
Mingyang Yu (MS at CMU → Sharpa)	Aug. 2024 – Mar. 2025
Dongsu Lee (MS at Soongsil U. → PhD at UT Austin)	Aug. 2024 – Mar. 2025
Chuye Hong (Undergrad. at THU → PhD at Berkeley)	Summer 2024
Yuming Feng (Undergrad. at PKU → MS at Stanford)	Summer 2024
Yunzhe Zhang (Undergrad. at THU → PhD at NYU Courant)	Feb. 2024 – Feb. 2025
Ziang Cao (Undergrad. at U. of Pittsburgh → MS at Stanford)	Summer 2023

Academic Service

Reviewer: RA-Letters, CoRL, ICRA, IROS, WAFR, MRS, ICLR, NeurIPS, ICML, AISTATS
Program Committee: NeurIPS 2023-2024 Workshop on Foundation Models and Decision Making (FMDM), CoRL 2023 Workshop on Language and Robot Learning (LangRob), Machine Learning for Autonomous Driving (ML4AD) Symposium 2023, NeurIPS 2022 Workshop on ML4AD
Organizer: ICRA 2024 RoboDrive Challenge, ICRA 2023 RoboDepth Challenge